

JAMES JAW

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EDUCATION

University of Wisconsin-Madison <i>M.S. in Computer Science</i>	Sep 2023 – Dec 2025 <i>GPA: 3.9</i>
National Tsing Hua University <i>Post-Baccalaureate in Computer Science</i>	Sep 2020 – Jan 2022 <i>GPA: 3.9</i>
Tunghai University <i>B.B.A. in Business Administration</i>	Sep 2014 – Jul 2018

SKILLS

Languages: Python, Golang, Java, C++, JavaScript, TypeScript, HTML/CSS
Frameworks: Django, Spring, React, Node.js, gRPC, REST APIs, PostgreSQL, MySQL
Infrastructure: Docker, Kubernetes, GitHub Actions, AWS, Linux

EXPERIENCE

Open Source – Contributor KubeRay, Apache Airflow – <i>Golang, Python, TypeScript, Docker, Kubernetes</i>	Feb 2026 – Present
– Diagnosed a KubeRay worker-startup bottleneck: right-sized the wait-gcs-ready init container's CPU/memory to match the Ray container, cutting up to 55s of autoscaling latency per new worker – Resolved a flaky e2e test caused by a concurrent-update race condition and modernized lint rules across the operator codebase; both merged upstream – Contributed 5 merged PRs to Apache Airflow, including a React rendering fix for duplicate-key list reconciliation in the web UI and a declarative YAML migration of provider connection metadata across 3 provider packages	
Digital Aid Seattle – Software Engineer Scheduling Platform – <i>TypeScript, Next.js, Supabase (Postgres/Auth/Edge Functions)</i>	May 2026 – Present
– Built a full-stack scheduling tool that automates student grouping via timezone-aware availability matching, replacing manual spreadsheet workflows for multiple instructors – Designed multi-tenant data isolation with Supabase Row Level Security, enabling instructors to independently manage class data on a shared platform	
IBM – Software Engineer Intern Middleware Modernization – <i>C++, Bash, Docker, Kubernetes, Argo CD, Linux, Grafana k6</i>	Jul 2025 – Aug 2025
– Built a C++ middleware integrating legacy CORBA APIs with modern deployment infrastructure, automating a previously manual validation workflow (~60% faster) – Migrated legacy C++ services to Kubernetes, replacing manual deployments with Blue-Green GitOps releases – Built team-friendly k6 gRPC load-testing tooling to simulate 1,000+ concurrent users and expose bottlenecks	

PROJECTS

Video Chat Platform – <i>Golang, Fiber, TypeScript, React, WebRTC, Postgres</i>	2026
– Shipped BuckyChat, a campus-only video chat platform for verified UW–Madison students, as the sole engineer from zero to production – Architected peer-to-peer WebRTC media over a lightweight Go signaling backend with Cloudflare TURN fallback; k6 load tests sustained 200 concurrent WebSocket sessions and 27,000+ connections at 100% handshake success – Enforced campus-only access via @wisc.edu domain allowlisting at signup and JWT-validated WebSocket sessions, with per-user rate limiting to curb abuse	
Parallel Computing Portfolio – <i>C++, OpenMP, CUDA</i>	2025
– Implemented a shared-memory tiled GEMM CUDA kernel (templated for int/float/double) and an optimized parallel reduction using first-add-during-load and sequential addressing; benchmarked against naive kernels across block sizes – Built a 3D gravitational N-body simulator with a leapfrog integrator; parallelized the $O(N^2)$ force loop with OpenMP and measured strong-scaling speedup up to N threads – Developed a 1D stencil kernel with halo-cell loading in shared memory, and an OpenMP task-parallel merge sort with a tuned serial cutoff	